

Chapter 1

Introduction

Foundation University currently manages several institutional services — clinic operations, guidance counselling, and biodegradable waste composting — through disconnected, largely manual processes. Clinic and counselling records are maintained separately with no shared data layer, appointments and referrals are tracked on paper, and medicine inventory is managed without integration to procurement. At the same time, the university has begun fabricating multiple units of a Biodegradable Waste Management (BMG) system — rotating drums that compost biodegradable waste such as food scraps, twigs, and leaves into usable fertilizer — but no digital system currently exists to record or monitor this process either.

Although these two domains, health/counselling services and facilities/sustainability operations, serve entirely different institutional functions, they share the same underlying structural problem: critical institutional operations are undocumented, siloed, or dependent on informal record-keeping, which limits the university's ability to monitor performance, demonstrate regulatory compliance, and make data-driven decisions.

On the health and counselling side, fragmentation leads to duplicated data entry, inconsistent record quality, and loss of holistic student and employee visibility. This mirrors documented cases at a much larger scale — such as Italy's disconnected regional electronic health record systems, which generate an estimated €3.3 billion annually in avoidable interregional healthcare costs due to the inability to share patient records across jurisdictions (Fedeli et al., 2025). While Foundation University operates at an institutional rather than national scale, the underlying dynamic is structurally identical: when services cannot share data, administrative burden escalates and holistic care becomes difficult to deliver.

On the facilities side, Republic Act (RA) 9003, the Ecological Solid Waste Management Act of 2000, mandates that institutions adopt systematic, ecological solid waste management programs that include composting of biodegradable materials (Official Gazette of the Republic of the Philippines, 2001). Despite this legislation, institutional composting programs in the Philippines often remain informal, undocumented, and difficult to evaluate (Commission on Audit [COA], 2023). Rotary drum composting, the method used in Foundation University's BMG units, is recognized in the literature as an effective way to accelerate decomposition through continuous tumbling, but actual decomposition duration is highly sensitive to drum design, waste composition, moisture, and operating conditions — meaning a generic baseline (e.g., a 21-day assumption) cannot be reliably applied to newer, more compact units without real operating data to confirm it (Michel et al., 2022; Sharma et al., 2024; Almulla et al., 2024). Without a tracking system, the university cannot determine whether its new units are actually more efficient, compare performance across units, or understand how waste type affects processing time. This mirrors a recurring gap identified in studies of Philippine schools and universities, where composting is practiced but structured monitoring remains absent, limiting programs' ability to measure effectiveness or justify further investment (Carale et al., 2025; Dinglasan & Duenas, 2024).

Comparable institutions have addressed similar monitoring gaps, in both domains, by developing purpose-built digital systems. In healthcare contexts, integrated platforms that consolidate previously siloed data flows under a single, governed system have produced measurable gains in data accuracy and administrative efficiency. In waste management contexts, Jammoul et al. (2023) developed ChemSorter, a web-based chemical waste management system that enabled the University of Balamand to maintain real-time inventories and reduce data entry time, while Jakimiuk et al. (2023) found that a structured, control-based waste disposal system at the Warsaw University of Life Sciences reduced total waste generation and improved separation quality. These studies, though situated in different domains, demonstrate the same principle: structured, web-based digital interventions produce measurable improvements over informal, manual processes.

Guided by these findings, this study proposes SYNAPSE, a unified web-based platform consisting of three core modules: Clinic Core, Counselling Core, and Facilities/Sustainability Core (BMG Tracking), all governed under a single authentication and role-based access control (RBAC) system. While the Facilities/Sustainability module operates independently of the Clinic and Counselling modules and does not exchange data with them, all three modules share the same integration philosophy: replacing fragmented, informal institutional record-keeping with structured, centralized digital systems governed by one login, one audit trail, and one quality standard. SYNAPSE's Clinic and Counselling modules consolidate patient records, appointments, referrals, and medicine inventory for both students and employees under a shared database, while its Facilities/Sustainability module serves as a digital logbook and analytics tool for the university's BMG composting units. In doing so, SYNAPSE operationalizes the policy intent of the WHO Global Strategy on Digital Health (WHO, 2021; WHO, 2025) and RA 9003 alike, and provides Foundation University with real-world operational data across both domains that prior fragmented or lab-controlled approaches could not offer.

1.1 Rationale of the Study

The rationale of this study rests on four intersecting concerns: regulatory and policy compliance, operational efficiency, data security and governance, and the wider gap in evidence-based institutional research.

First, from a regulatory standpoint, both domains addressed by SYNAPSE are governed by clear national policy. On the health side, the Philippines operationalizes data governance obligations through Republic Act No. 10173 (the Data Privacy Act of 2012) and NPC Circular No. 2023-06, which mandate organizational, physical, and technical safeguards for sensitive personal data, including health and counselling records (National Privacy Commission, 2023). On the facilities side, RA 9003 requires institutions, including higher education institutions, to implement ecological solid waste management programs that include composting at source (Official Gazette of the Republic of the Philippines, 2001), yet national monitoring by the Commission on Audit shows uneven implementation, with only 245 of 11,637 established Materials Recovery Facilities operational nationwide as of 2021 (COA, 2023). A unified digital system that documents both health service delivery and composting operations strengthens the university's compliance posture across both regulatory frameworks simultaneously.

Second, from an operational standpoint, both domains face a structurally similar problem: growing operational complexity that informal record-keeping cannot adequately address. On the health side, the addition of employee records alongside student records, combined with cross-referrals between clinic and counselling staff, increases the volume and sensitivity of data that must be tracked. On the facilities side, the deployment of multiple BMG units introduces a management challenge that manual logbooks cannot scale to meet, increasing the risk of data loss, transcription errors, and inconsistent reporting (Jakimiuk et al., 2023). A single platform with role-based access, automated calculations, and centralized reporting directly addresses these operational risks across both domains.

Third, from a data governance standpoint, healthcare data is classified as sensitive personal information under the Data Privacy Act, requiring the highest level of protection (National Privacy Commission, 2023). Hybrid cryptographic access control, combining role-based authorization with application-layer encryption, has been shown to provide significantly stronger protection than RBAC alone (Chinnasamy & Deepalakshmi, 2022). SYNAPSE's shared RBAC and audit-logging architecture extends this same governance discipline to its Facilities/Sustainability module, ensuring consistent accountability across all three modules even though their data domains remain separate.

Fourth, the rationale extends to the academic and research community. Existing studies on rotary drum composting are largely conducted in laboratory or controlled school settings and are not always representative of day-to-day institutional operating conditions (Michel et al., 2022; Almulla et al., 2024; Sharma et al., 2024). Similarly, health information system studies often examine fragmentation and access governance in isolation from real institutional deployment. By collecting real-world operating data across all three of its modules, SYNAPSE contributes datasets that address both gaps and can inform future research on decentralized institutional service management in Philippine higher education settings.

1.2 Objectives of the Study

1.2.1 General Objective

The general objective of this study is to develop SYNAPSE, a unified web-based platform that consolidates Foundation University's clinic, counselling, and facilities/composting operations under a single access-controlled system, supporting data-driven institutional decision-making across health services and sustainability operations.

1.2.2 Specific Objectives

Clinic and Counselling Module:

1. Develop clinic and counselling record modules under a shared database and unified role-based access control (RBAC) system.
2. Implement a QR-based referral verification feature that generates a unique, scannable code for each referral or excuse slip, allowing clinic and counselling staff to instantly verify authenticity and view referral details.

3. Implement an RFID-based check-in feature that allows clinic staff to scan a student's or employee's institutional ID to retrieve their record, reducing manual lookup and registration errors.
4. Implement a medicine inventory module integrated with the procurement/purchasing process, enabling automated reorder triggers based on stock thresholds and expiration tracking.
5. Extend system scope to include employee health and counselling records, with employee data gathered in coordination with the Human Resources (HR) department to support a complete institutional user base.

Facilities/Sustainability Module (BMG):

6. Develop a web-based system that allows authorized users to record the daily input weight or volume of biodegradable waste loaded into each BMG unit.
7. Implement a process-tracking feature that monitors and calculates the exact duration, in days, required for each batch to fully decompose into fertilizer.
8. Implement an output-tracking feature that records the final weight or volume of fertilizer produced per batch, accounting for expected yield loss during decomposition.
9. Design the system to support waste categorization (e.g., pure food waste, twigs and leaves, or mixed waste) to help explain variance in decomposition duration across batches.
10. Create a scalable dashboard architecture that allows individual BMG units to be added, edited, or archived as the university's composting program expands or contracts.
11. Develop reporting and analytics features that present input, duration, and yield data in a format usable by the university's facilities or environmental management office for operational decision-making.

System-Wide:

12. Evaluate all three modules' usability and functionality through User Acceptance Testing (UAT) with representative end users, using the ISO/IEC 25010 quality model as a shared reference framework.

1.3 Significance of the Study

This study is significant to the following stakeholders:

Clinic Staff (Clinic Administrator, Doctor, Nurse). SYNAPSE consolidates patient records, appointment scheduling, and medicine inventory under one system, replacing paper ledgers and disconnected spreadsheets, and reducing service delays caused by manual data entry.

Guidance Counsellors. The system provides a structured, encrypted platform for counselling appointments and referrals, with a defined referral bridge to the clinic for cases requiring medical follow-up or escalation.

Students and Employees. Both groups gain a single point of access for booking appointments, submitting referrals, and receiving timely reminders, with employee inclusion ensuring the system serves the university's complete population rather than students alone.

University Administration. Consolidated, real-world data from both the health/counselling and facilities domains supports informed decisions on resource allocation, staffing, program expansion, and compliance reporting under both the Data Privacy Act and RA 9003.

Facilities and Environmental Management Staff. The Facilities/Sustainability module provides a centralized, structured way to monitor BMG unit performance without relying on informal or manual tracking, enabling faster identification of underperforming units, unusually slow batches, and low-yield waste categories.

Facilities Staff Operating the BMG Units. These staff gain a simple, structured interface for logging daily input and output data, removing the need to maintain separate paper or spreadsheet records.

University Sustainability Program. The platform provides a documented, data-backed understanding of how effectively biodegradable waste is converted into usable fertilizer, supporting environmental reporting, accreditation requirements, and continuous improvement of the university's green initiatives.

Future Researchers. SYNAPSE produces real-world datasets across two otherwise unrelated institutional domains, health service delivery and composting operations, generated under actual operating conditions. This addresses gaps left by prior lab-controlled and fragmented-system studies, and may serve as a reference for future research on institutional digital health systems and decentralized biodegradable waste management in Philippine higher education settings.

Local Government and the Wider Community. Although SYNAPSE's primary scope is Foundation University, its methodology, architecture, and lessons learned, spanning both health service consolidation and composting monitoring, may inform other schools, local government units, and small institutions in Negros Oriental and beyond.

1.4 Scope and Limitations of the Study

1.4.1 Scope

The scope of this study covers the design, development, and evaluation of SYNAPSE, a unified web-based platform consisting of three integrated modules, Clinic Core, Counselling Core, and Facilities/Sustainability Core (BMG Tracking), all governed under a single authentication and role-based access control (RBAC) system.

Clinic Module

- Digitized patient records for students and employees, replacing paper ledgers and disconnected spreadsheets
- QR-based referral/excuse slip verification, generating a unique scannable code per referral for authenticity checks
- RFID-based check-in for retrieving student/employee records at the point of service
- Medicine inventory management integrated with the procurement process, including automated low-stock alerts, expiration monitoring, and reorder requests

- Basic clinical encounter logging for walk-in and scheduled visits

Counselling Module

- Digitized counselling appointment scheduling with automated reminders and no-show tracking
- A bidirectional referral system connecting clinic and counselling staff, including escalation pathways for crisis-level cases
- Partitioned, encrypted counselling records accessible only to authorized counselling staff

Employee Scope (HR Integration)

- Extension of clinic and counselling record-keeping to cover university employees, in addition to students
- Employee data gathered in coordination with the Human Resources (HR) department to populate and maintain employee user profiles

Facilities/Sustainability Module (BMG Tracking)

- Input tracking: daily recording of biodegradable waste weight or volume loaded per BMG unit
- Process tracking: calculation and monitoring of decomposition duration, in days, per batch, using the difference between batch start and completion dates
- Output tracking: recording of final fertilizer weight or volume per completed batch, including auto-calculated yield percentage (output divided by input)
- Waste categorization: tagging of batches by type (e.g., food waste, twigs and leaves, mixed) to support comparative analysis
- Scalable multi-unit dashboard allowing BMG units to be added, edited, or archived as the physical composting program changes
- Reporting and analytics consolidating input, duration, and yield data across units and time periods, with PDF/CSV export for institutional review

System-Wide

- Role-based access control across all modules (System Administrator, Clinic Staff, Guidance Counsellor, Facilities/Composting Staff, Report Viewer, and Employee/Student user roles)
- A shared, write-once audit trail logging access and data-entry actions across all three modules
- Usability and functional suitability evaluation using the ISO/IEC 25010 quality model, validated through User Acceptance Testing (UAT) with representative end users from each module

1.4.2 Limitations

Clinic & Counselling Limitations

- Clinical & Medical: The system excludes e-prescriptions, medicine dispensing, telemedicine, billing, and integration with any national health database
- Guidance & Academic: Standardized testing within the counselling module is limited to developmental profiling and excludes clinical psychiatric screening or diagnosis

- Hardware ownership: The development team provides the QR/RFID scanning hardware and application logic; the university is responsible for issuing and maintaining the RFID-enabled institutional ID cards used for check-in
- Procurement scope: The medicine inventory module tracks stock and triggers reorder requests but does not process actual purchase transactions, vendor payments, or supplier contracts, which remain the responsibility of the university's procurement office

Facilities/Sustainability (BMG) Limitations

- Hardware scope: The study does not include the fabrication, physical construction, or mechanical operation of the BMG units themselves; the system is a software solution only
- Data entry mode: The BMG module does not include automated sensors, IoT hardware, or scanning devices for composting data; all input, batch start/end dates, and output weight are entered manually by authorized personnel using a weighing scale already present at each BMG unit location
- Scientific interpretation: The system does not perform scientific interpretation of composting data, such as determining optimal waste ratios or improving decomposition chemistry; this remains the responsibility of the university's facilities or environmental management office
- Logistics scope: The system does not manage waste collection logistics, transportation, or sourcing of biodegradable waste prior to loading into a BMG unit

System-Wide Limitations

- Module independence: The Facilities/Sustainability (BMG) module operates independently of the Clinic and Counselling modules and does not exchange data with them; integration is limited to shared login, RBAC governance, and institutional audit logging
- Operating environment: System evaluation is limited to Foundation University's clinic, counselling, and BMG operations during the study period; generalization to other institutions, climates, or drum designs is outside the scope of this study
- Connectivity and infrastructure: System availability is dependent on the university's server and internet infrastructure; periods of downtime or loss of connectivity may interrupt data entry, though locally cached entries will synchronize upon reconnection

1.5 System Workflow (User–System Interaction)

1.5.1 Clinic and Counselling Workflow

Check-In (RFID): Clinic staff scan a student's or employee's institutional ID at the point of service. The system retrieves the corresponding record and logs the encounter timestamp.

Referral Verification (QR): When a referral or excuse slip is issued, either from clinic to counselling or vice versa, the system generates a unique QR code tied to that referral record. The receiving party scans the code to instantly verify authenticity and view referral details, replacing unverifiable paper slips.

Appointment Scheduling: Students and employees book clinic or counselling appointments through the system, which sends automated reminders and records no-shows for follow-up.

Medicine Inventory: Clinic staff log medicine stock levels; the system monitors thresholds and expiration dates, automatically flagging items for reorder and routing requests to the appropriate procurement personnel.

1.5.2 Facilities/Sustainability (BMG) Workflow

This workflow follows an Input → Process → Output pattern, in which each step is initiated by a user action and finalized by a system computation or validation. Each BMG unit is represented on the dashboard as a drum whose status alternates between "Idle" (no active batch) and "Processing" (a batch is currently being composted).

Input Tracking: Facilities staff weigh the collected biodegradable waste on a physical scale, select the specific composting drum and waste category in the system, and enter the total weight in kilograms. The system timestamps this as the batch's start date, logs the weight and category linked to that drum's ID, and updates the drum's status to "Processing."

Process Tracking: Staff monitor the physical drum over the following days or weeks. Once the waste has visually decomposed into mature fertilizer, staff select the active batch and click "Mark as Completed." The system captures the completion date and automatically calculates Total Duration (Completion Date – Start Date), then resets the drum's status to "Idle."

Output Tracking (Yield Analytics): Staff weigh the final harvested fertilizer and enter this weight into the system. The system calculates Yield Percentage ($\text{Output Weight} \div \text{Input Weight} \times 100$) and Mass Reduction ($100\% - \text{Yield Percentage}$). A validation rule blocks submission if the output weight exceeds the input weight, since this is scientifically impossible; this is enforced both at the application layer and via a database-level constraint.

Together, these three steps form a complete audit trail per batch, allowing the university to scientifically compare new, compact drums against older baseline assumptions and to determine exactly how efficiently campus waste is being converted into usable fertilizer.